Modèle TK IMD

v1.1

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1 Toxicokinetics of Epoxiconazole

1.1 Results of the experiment

1.1.1 Data importation

Notations :

- `Date_start` : Start of the experiement (t0)
- `Molecule` : Substance to which the earthworm is exposed
- `Dose` : Exposure dose (mg/kg)
- `ID_recipient` : Recipient in which the soil was prepared
- `ID_camp` : ID of the earthworm campagne and generation (see XXX for more information)
- `ID` : ID of the earthworm
- `Nb_rep` : Identification number of the replicate
- `Time_point` : Time point to which the earthworm is dedicated
- `Phase` : Uptake or Elimination phase
- `Time_introduction_D0` : Time of the day when the earthworm was introduced into his contaminated cosm
- `w0` : Body weight at t0
- `w_Tp` : Body weight at the given time point
- `w_frozen` : Body weight before congelation
- `C_worm` : Measured concentration of the substance in the earthworm
- `C_soil` : Measured concentration of the substance in the soil at the given time point

1.1.2 Initial state

1.1.3 Growth curves

```
       ID            w1           Molecule               w2
 5      :  2   Min.   :0.5133   Length:128         Min.   :0.3968
 6      :  2   1st Qu.:0.6015   Class :character   1st Qu.:0.6576
 7      :  2   Median :0.6821   Mode  :character   Median :0.7644
 8      :  2   Mean   :0.7106                      Mean   :0.7739
 13     :  2   3rd Qu.:0.8147                      3rd Qu.:0.8924
 14     :  2   Max.   :0.9937                      Max.   :1.2453
```

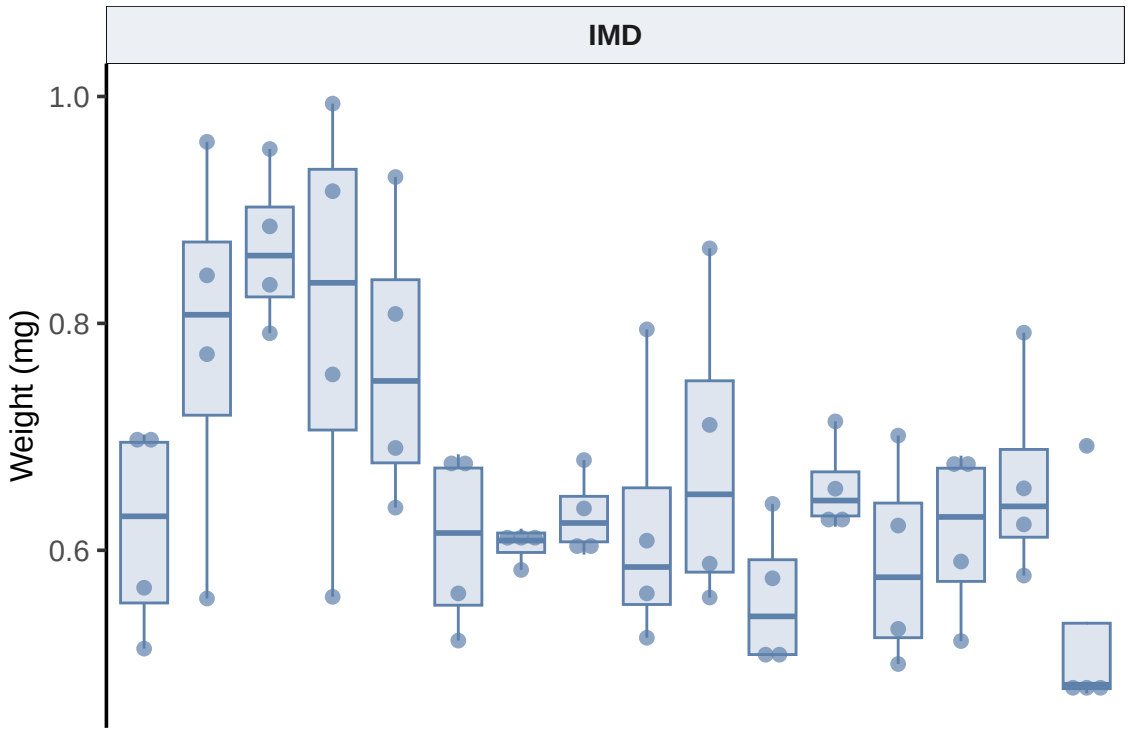


Figure 1.1: Initial weights (with gut content) of earthworms.

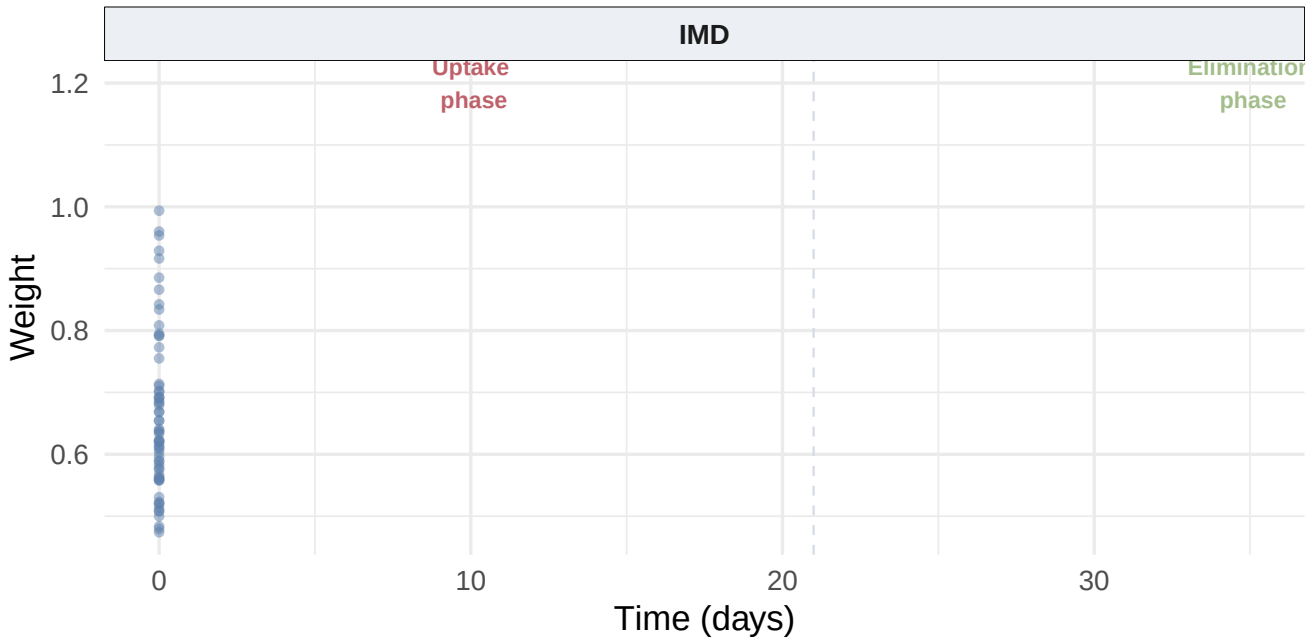


Figure 1.2: Evolution of earthworms body weight (with gut content) throughout the experiment.

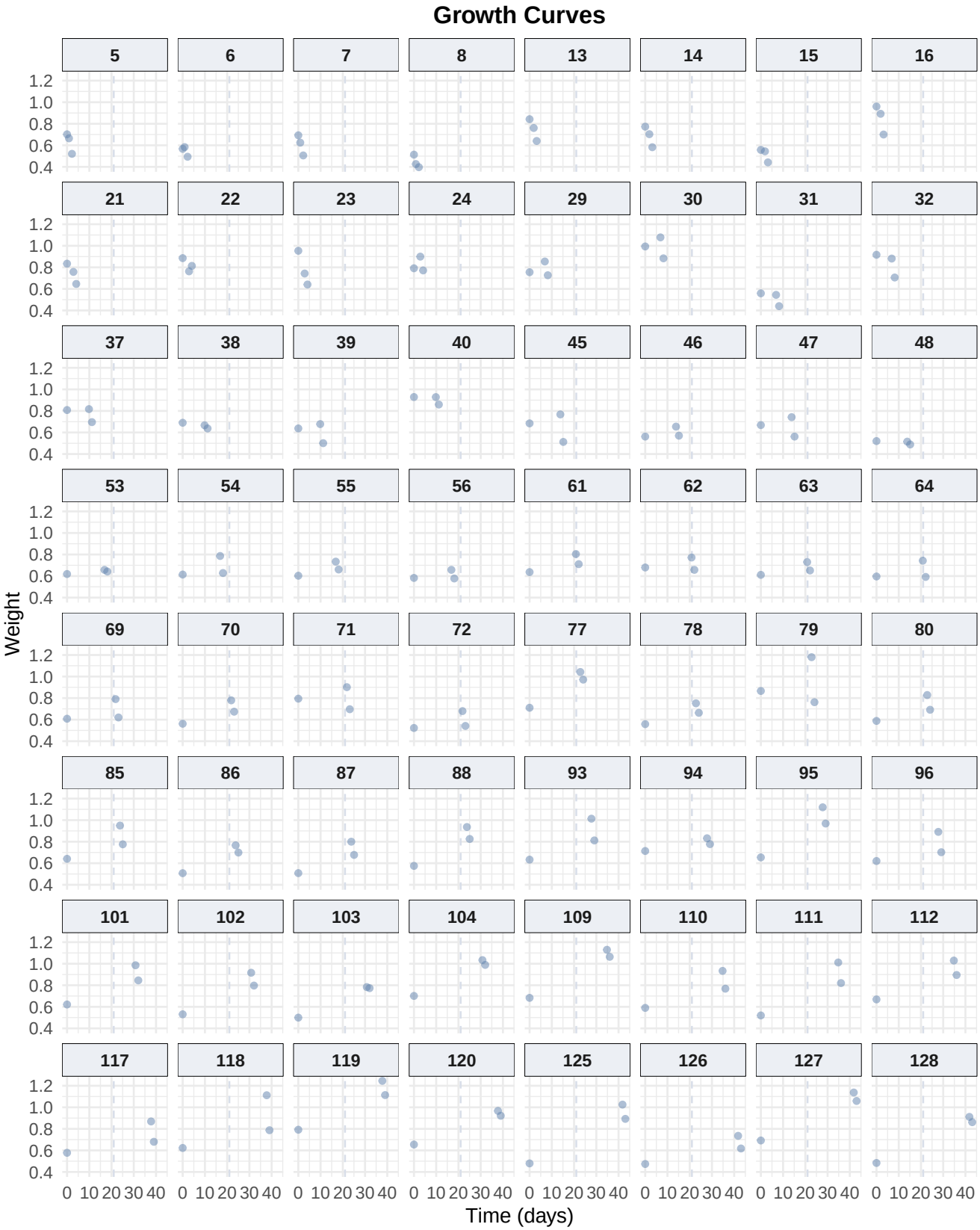
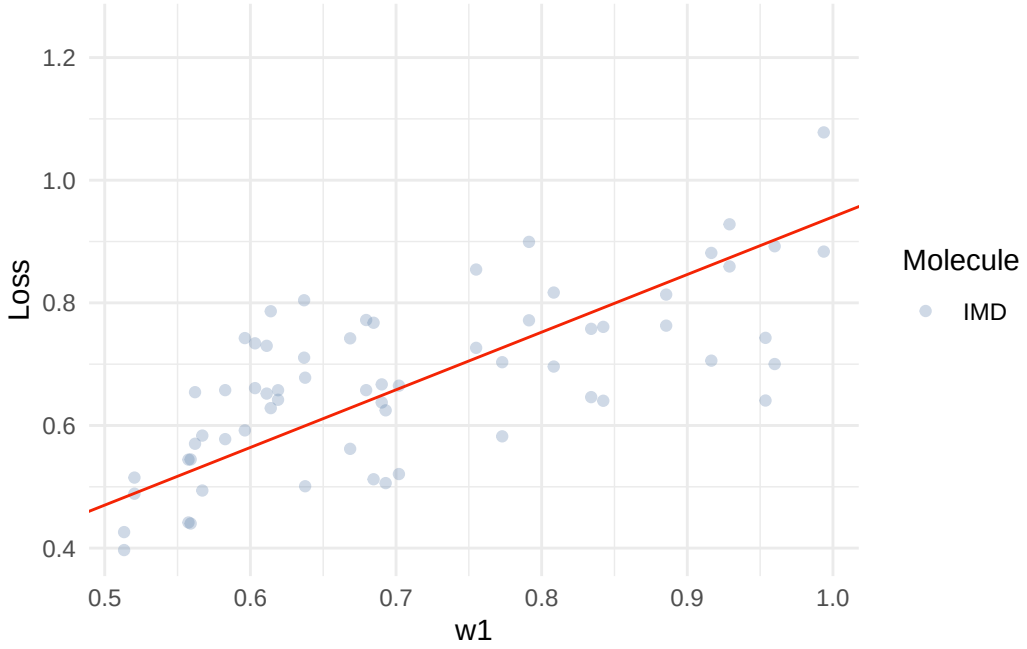


Figure 1.3: Evolution of earthworms body weight before and after fasting

```
 (Other):116   NA's   :64
    Per_loss            Loss
 Min.   :-28.062   Min.   :0.3968
 1st Qu.: -6.409   1st Qu.:0.6576
 Median :  3.561   Median :0.7644
 Mean   :  3.993   Mean   :0.7739
 3rd Qu.: 14.397   3rd Qu.:0.8924
 Max.   : 32.820   Max.   :1.2453
 NA's   :64


Call:
lm(formula = Loss ~ w1 + 0, data = df_TK_fasting)

Residuals:
     Min       1Q   Median       3Q      Max
-0.25601 -0.06459  0.01838  0.07759  0.20899

Coefficients:
   Estimate Std. Error t value Pr(>|t|)
w1  0.94024    0.01823   51.59   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1056 on 63 degrees of freedom
  (64 observations deleted due to missingness)
Multiple R-squared:  0.9769,    Adjusted R-squared:  0.9765
F-statistic:  2662 on 1 and 63 DF,  p-value: < 2.2e-16

Warning: Removed 64 rows containing missing values or values outside the scale range
(`geom_point()`).
```



!!!!!!!! Growth rates !!!!!

1.1.4 Measured internal concentrations

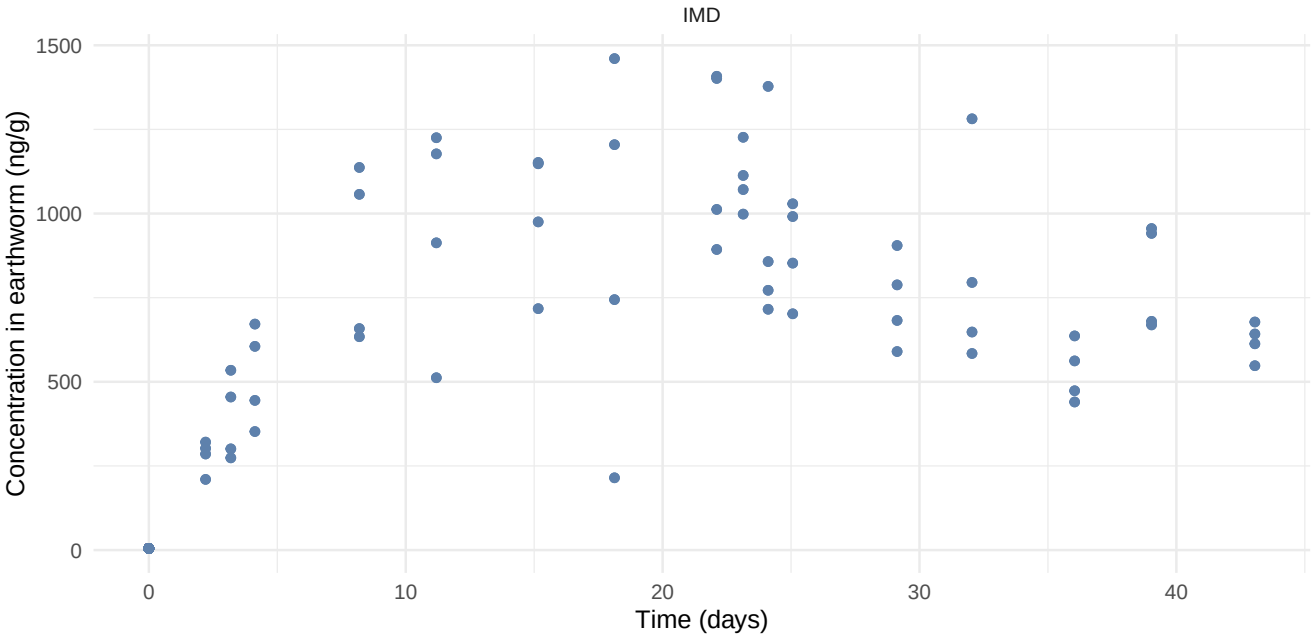


Figure 1.4: Measurements of IMD concentration in earthworms exposed to 10 mg/kg of IMD and related weight measurement of earthworms. Points represent the experimental data, the blue curve and the transparent blue band are produced with geom_smooth.

1.2 Modeling of the toxicokinetics

The first step is to retrieve the information of the MCMC simulation experiments.

The inputs :

- Input files must a name like "xxxxx.1.in"
- Output files must a name like "xxxxx.1.out"

We can retrieve different information from the model :

- The number of chains used : 3
- The number of parameters : 6
- The number of experiments : 64
- The number of last iterations used to describe the posterior distribution : 10^4
- The number of last iterations used to create the `tab_setpoint.out` file : 333

Warning : Run the command `./mcsim.m_TK_IMD TK_IMD_Setpoint.in` in the terminal while being in the folder `mod/TK_IMD` to create the `Setpoints.out` now. Do the same with `./mcsim.m_TK_IMD TK_IMD_Setpoint_ind.in` to create the `Setpoints_ind.out` and with `./mcsim.m_TK_IMD TK_IMD_Setpoint_ind_fu` to create the `Setpoints_ind_full.out`.

1.2.1 Priors used for the model

```
# A tibble: 6 x 6
  Nom         Distribution     P1          P2      P3    P4
  <chr>       <chr>         <dbl>       <dbl>   <dbl> <dbl>
1 kuIMD       LogUniform   0.0001 100000      NA         NA
2 keIMD       LogUniform   0.0001 100000      NA         NA
3 a_growth    TruncNormal  0.005       0.005  -0.5        1
4 Vr_a_growth TruncNormal  0.0015      0.0015  0          1
5 Sigma_W     TruncNormal  0.03        0.01    0.0001     1
6 Sigma_CiIMD TruncNormal  3           3       1       1000
```

1.2.2 Chains

We can also take a look at the likelihood and the deviance ratio.

Correlations :

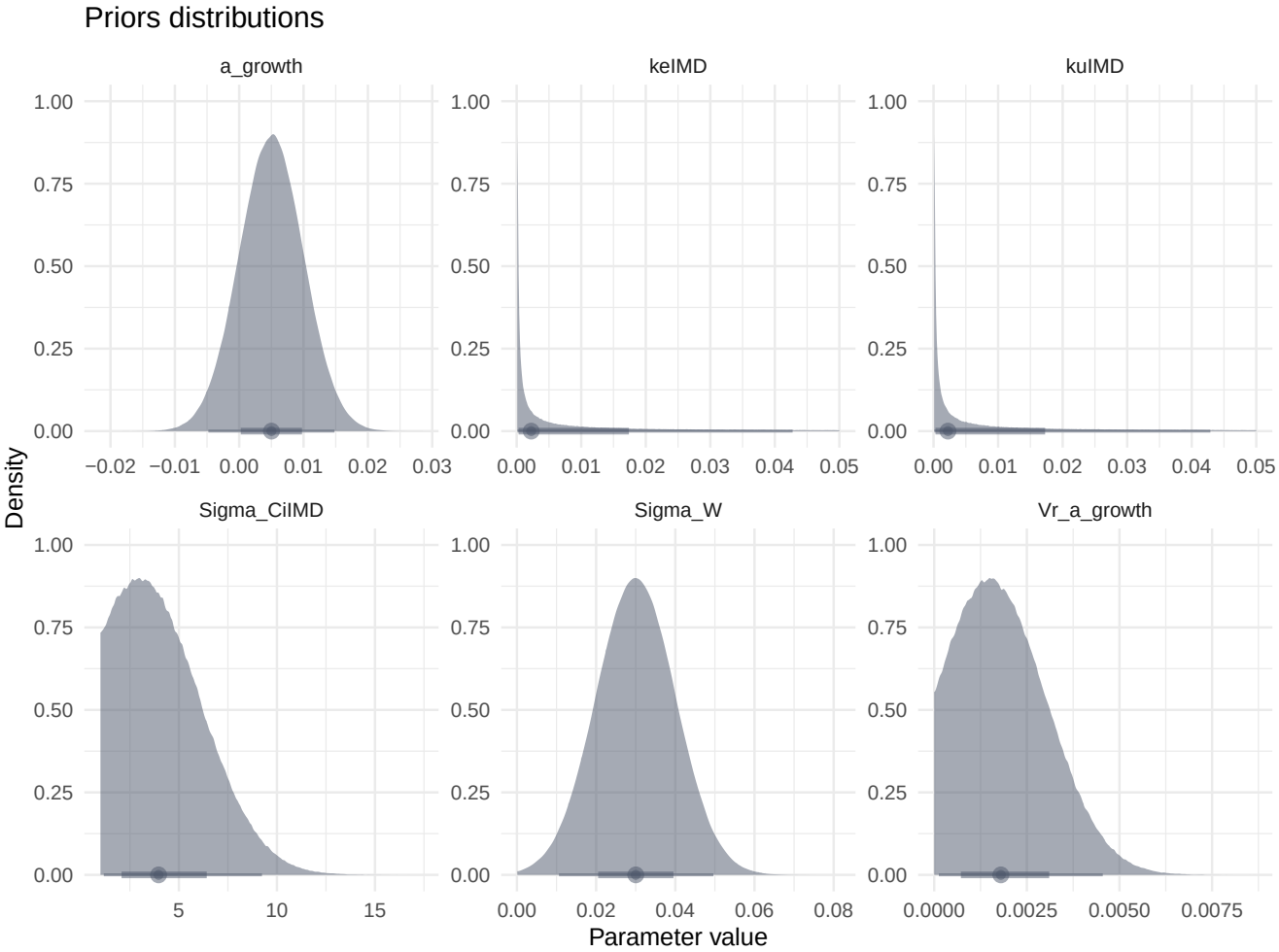


Figure 1.5: Distributions of the priors.

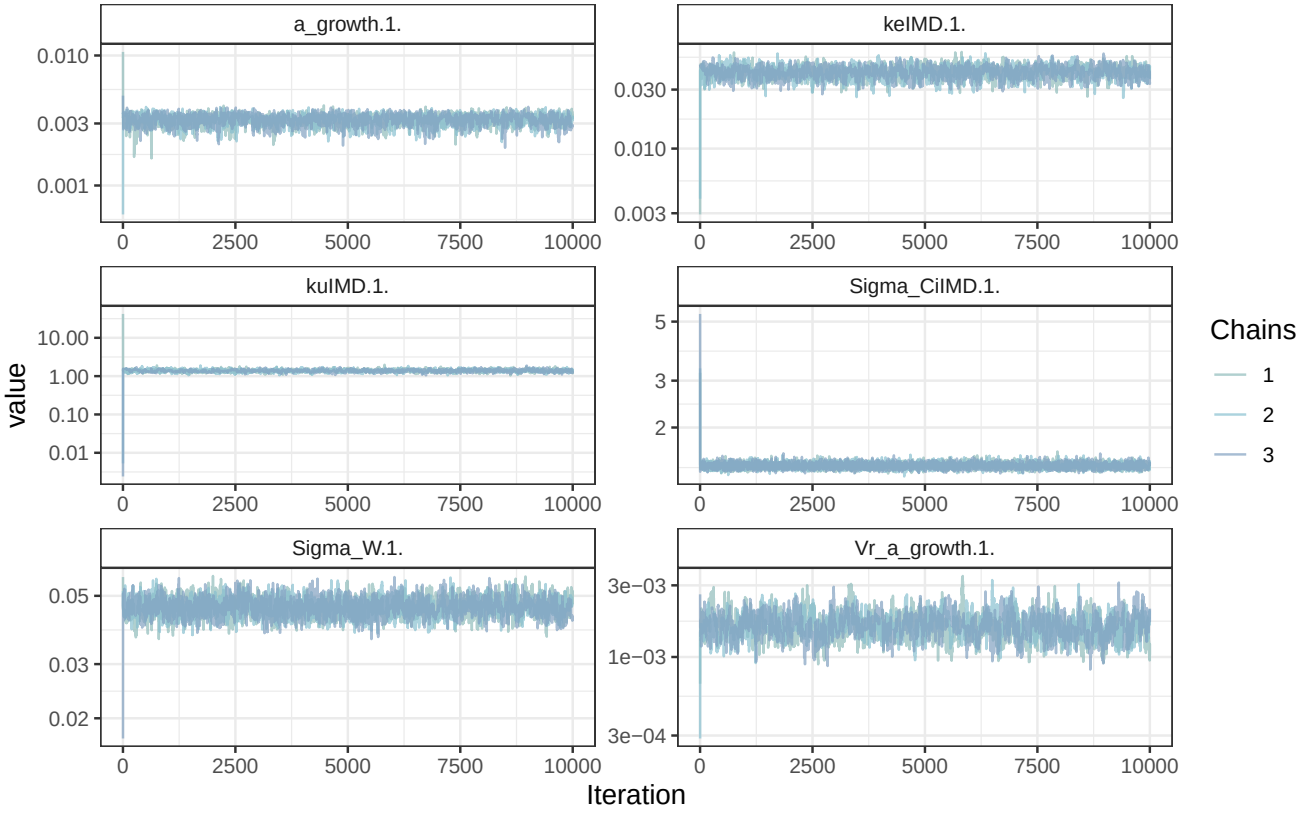


Figure 1.6: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

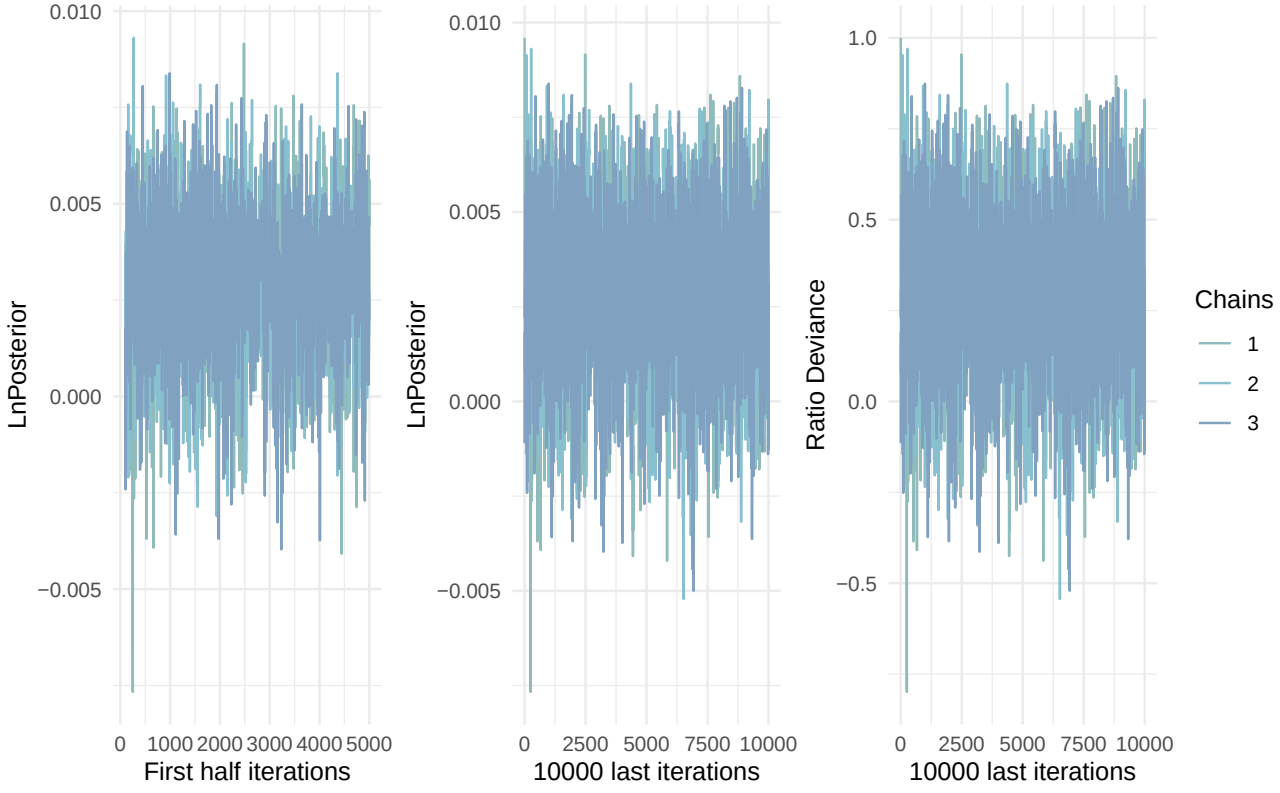


Figure 1.7: Evolution of likelihood and deviance ratio through the iterations.

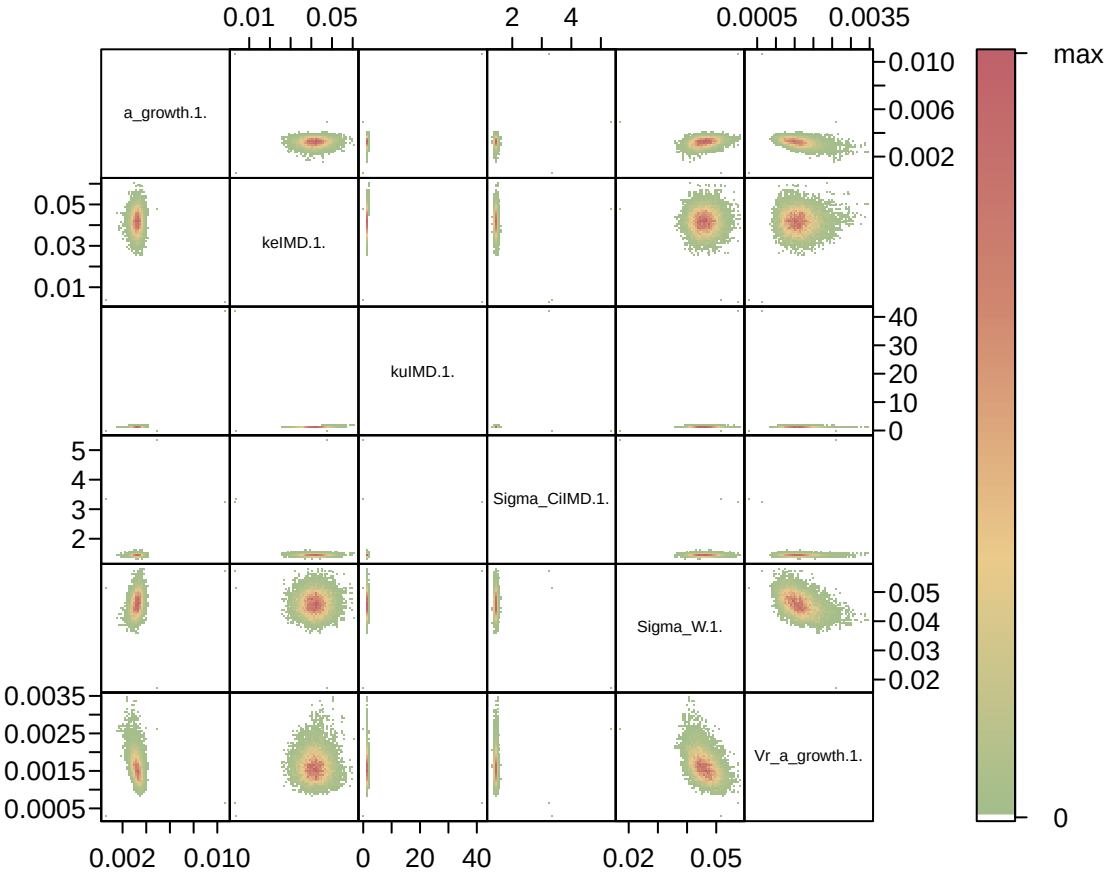


Figure 1.8: Parameter correlations.

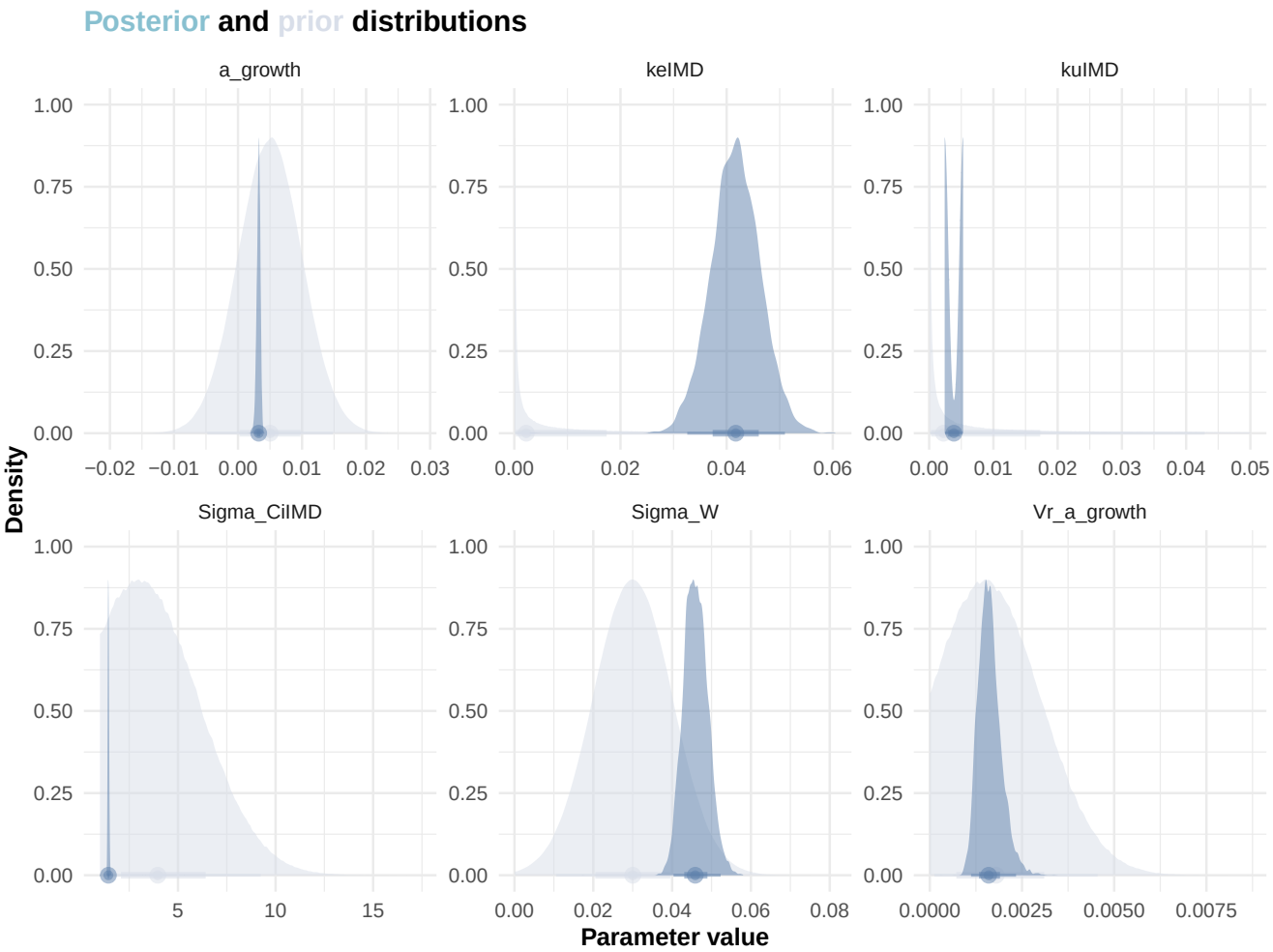


Figure 1.9: Distributions of the priors and the posteriors.

1.2.3 Parameter estimation results

```
# A tibble: 7 x 7
  kuIMD.1. keIMD.1. a_growth.1. Vr_a_growth.1. Sigma_W.1. Sigma_CiIMD.1.   BCF
     <dbl>    <dbl>       <dbl>          <dbl>      <dbl>          <dbl> <dbl>
1   1.3      0.04       0.0034         0.0013      0.046           1.4   0.12
2   1.2      0.033      0.0026         0.0011      0.04            1.4   0.049
3   1.6      0.051      0.0037         0.0023      0.052           1.5   0.16
4   0.0024   0.0029     0.0006         0.00029     0.017           1.3   0.04
5  42        0.06       0.011          0.0035      0.058           5.3   1.8
6   1.4      0.042      0.0032         0.0016      0.046           1.4   0.094
7   0.26     0.0046     0.00029        0.00031     0.0031          0.044 0.038
```

1.2.4 Comparison of observed vs. predicted data

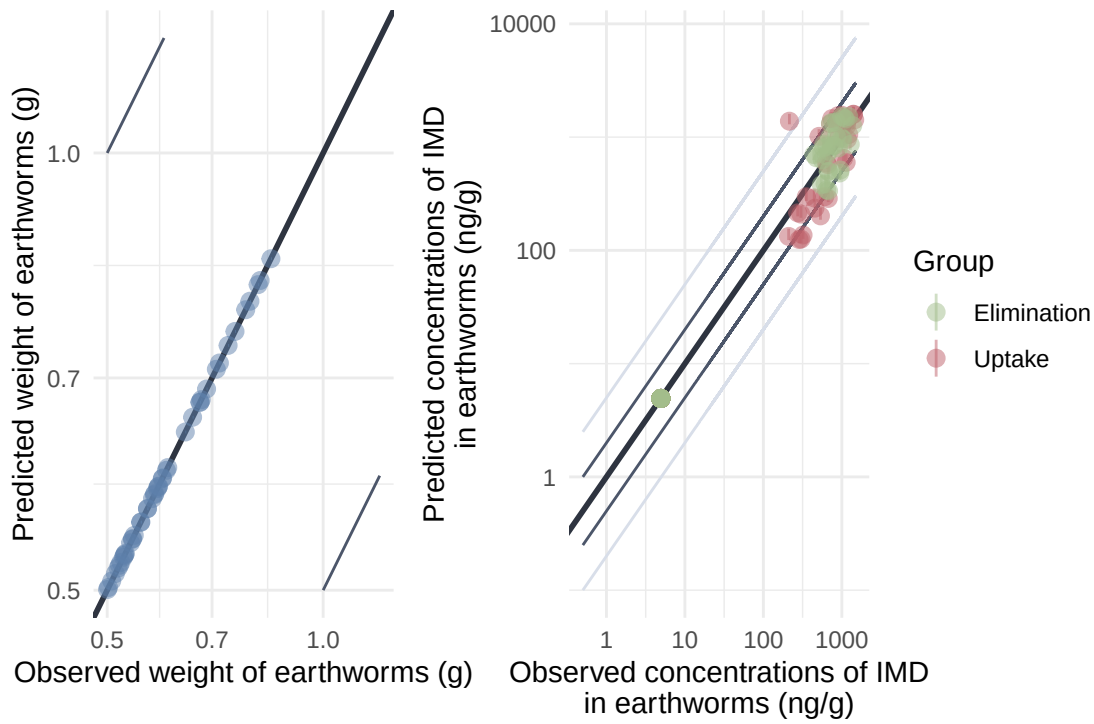


Figure 1.10: Results of the model fit on experimental data. Predicted values are represented with points with their respective credibility interval of 95%. Thin black and grey lines represent the 2-folds and 5-fold changes, respectively. The thicker black line represents the identity line.

```
For weights :

 Percentage of prediction in fold 2 =  100 %
 Percentage of prediction in fold 5 =  100 %
```

```
For internal concentrations :

 Percentage of prediction in fold 2 =  93.8 %
 Percentage of prediction in fold 5 =  99.2 %
```

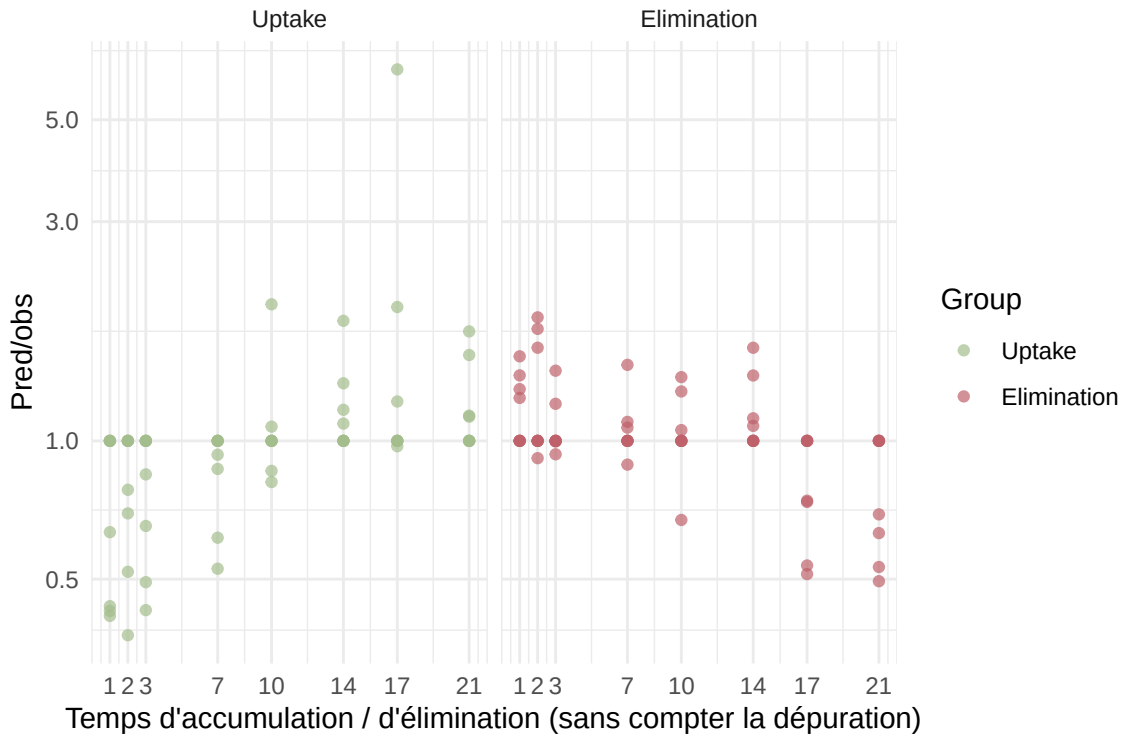


Figure 1.11: Residuals

1.2.5 Full individual trajectories

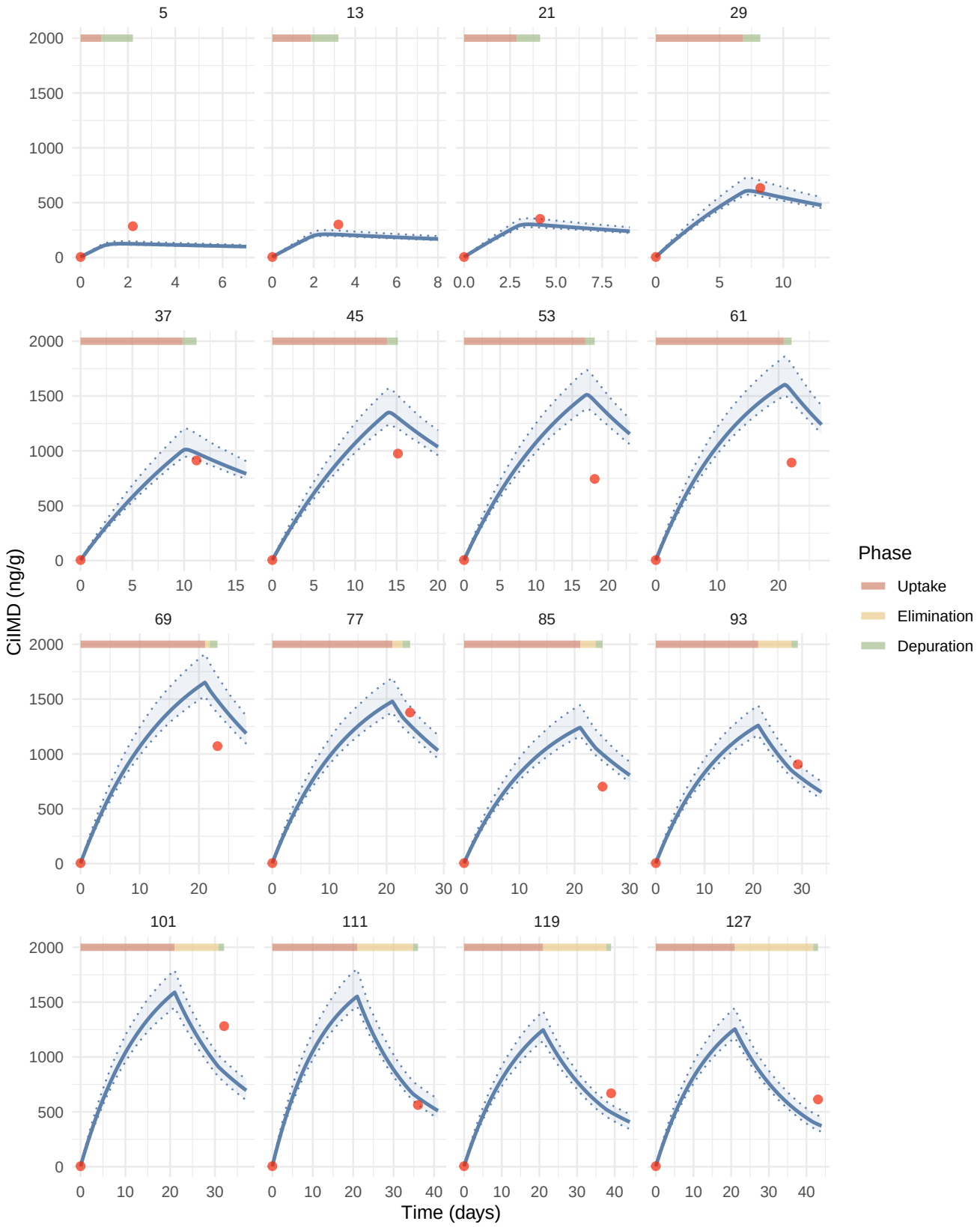


Figure 1.12: Estimated toxicokinetic of the IMD concentration in earthworms exposed to 1000 ng/g of IMD. Red points represent the experimental data, the blue curve represent the estimated toxicokinetic curve and the transparent blue band correspond to the credibility interval of the model.

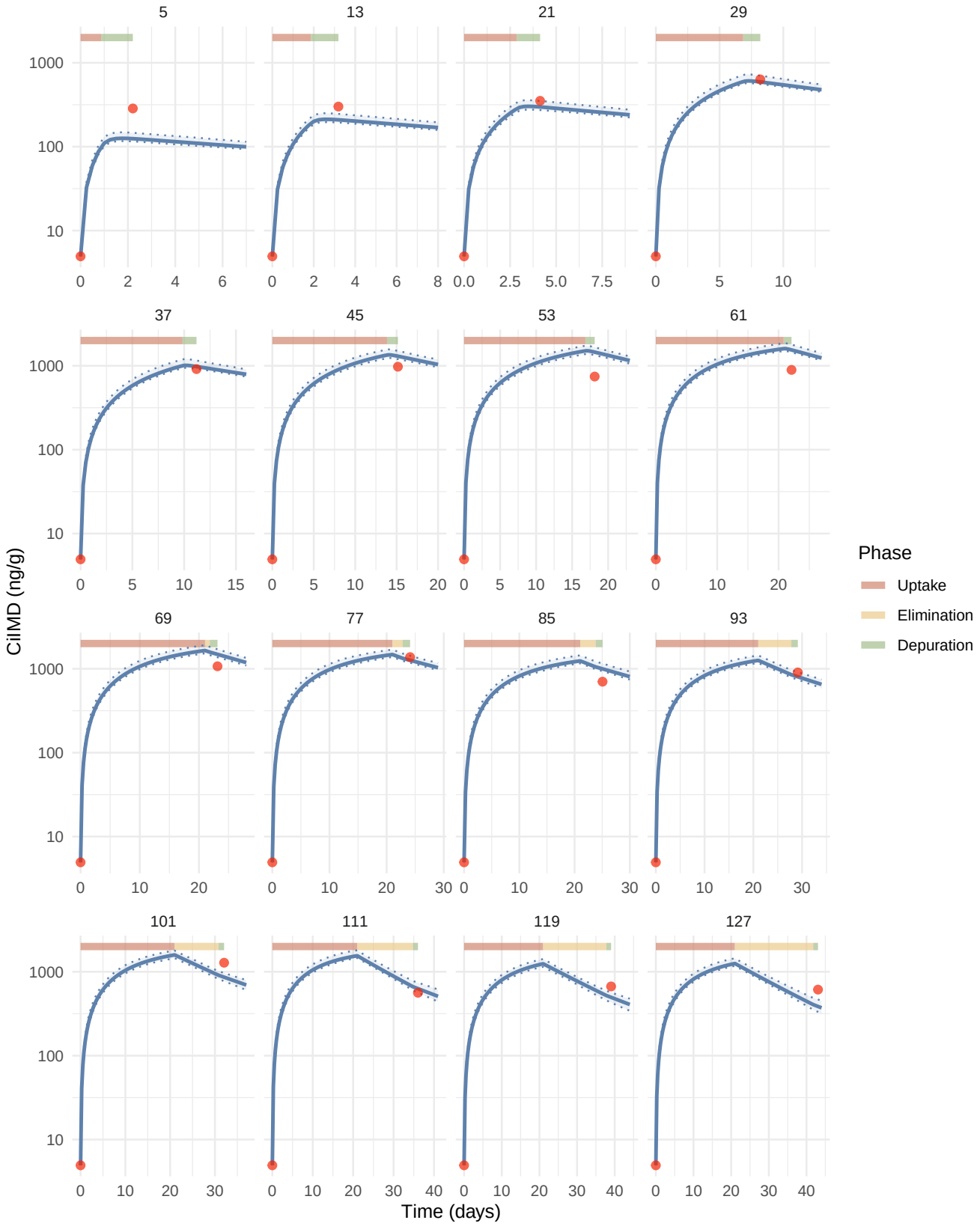


Figure 1.13: Estimated toxicokinetic of the IMD concentration in earthworms exposed to 1000 ng/g of IMD. Red points represent the experimental data, the blue curve represent the estimated toxicokinetic curve and the transparent blue band correspond to the credibility interval of the model.

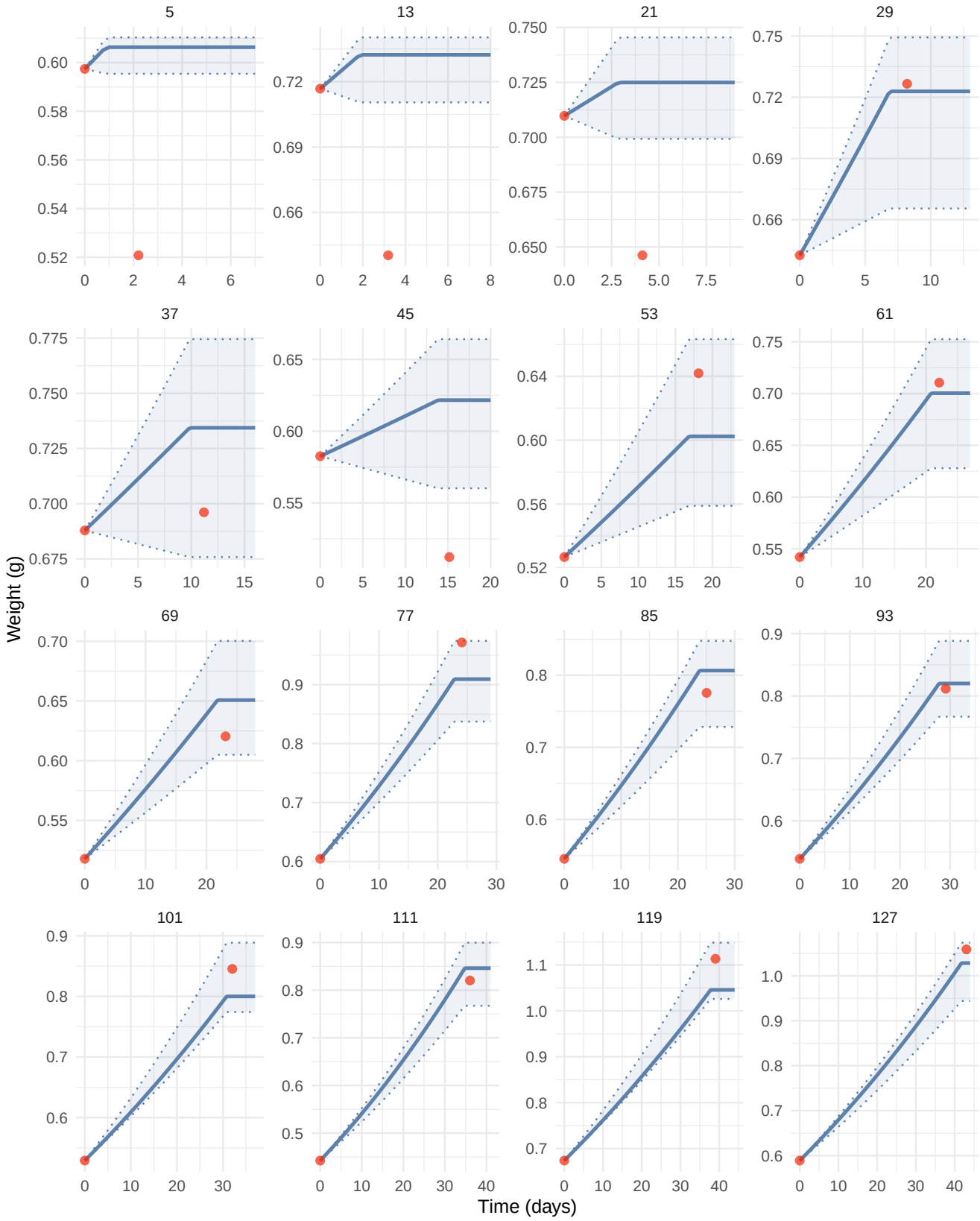


Figure 1.14: Weight predictions

1.2.6 Simulation of the toxicodynamic in earthworms throughout the exposition

The goal here is to recreate the concentration dynamic in the average earthworm during our exposition scheme but without the fasting phase that was required for chemical analysis and during which elimination was happening.

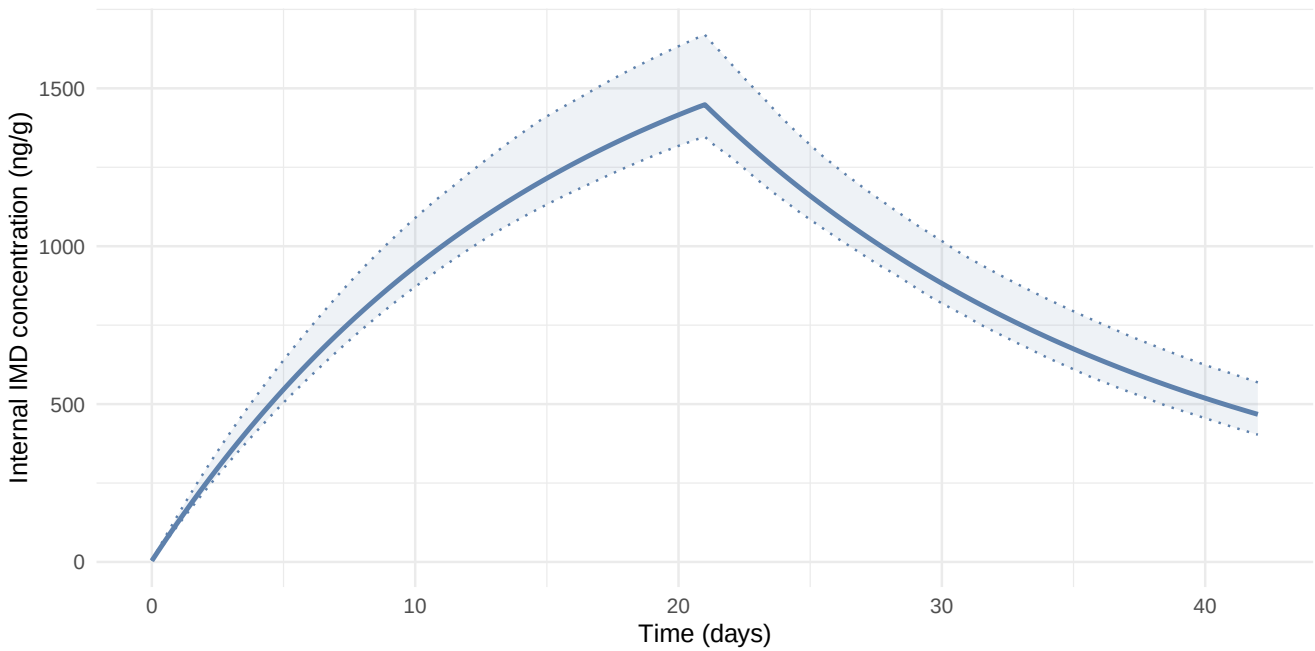


Figure 1.15: Estimated toxicokinetic of the IMD concentration in earthworms exposed to 0.1 mg/kg of IMD. Blue points represent the experimental data, the blue curve represents the estimated toxicokinetic curve and the transparent blue band corresponds to the credibility interval of the model.